Microscopes I

The history of microscopy

Ever since humans have studied the world around them, people have wanted to see things with greater detail than the naked eye permitted. An important event that advanced the development of cell theory was the invention of the microscope. There is some debate about specific dates and relative importance of contributions, but the following all played a part.

In Holland, near the end of the sixteenth century, both Hans Lippershey and the father and son team of Hans and Zacharias Janssen were credited with the invention of microscopes. Dutch clockmaker Anton van Leeuwenhoek (1632–1723) was also recognised with inventing the first practical microscope-like instrument. Leeuwenhoek's device was just two lenses which, he discovered by accident, dramatically increased magnification when placed one over the other. Robert Hooke (1635–1703) built another important microscope. Hooke's reflecting microscope was described in his 1665 book *Micrographia*, along with his illustrations of many of the objects he examined using it.

A microscope with two or more lenses in it is a *compound microscope* (*compound* to combine or put together). The word *lens* comes from the Latin word for lentil, which is the shape of a convex lens.

Marcello Malpighi (1628–1694) was an Italian physiologist who, in 1661, used a microscope to study blood circulation. In 1838 Matthias Schleiden, a German botanist, studied plant parts under a microscope and discovered that plant parts are made of cells. A year later, Theodor Schwann, a German zoologist, examined animal parts under a microscope and revealed that animal parts are also made of cells.

Around 1880 to 1890 Pasteur and Koch made many discoveries using microscopes. Louis Pasteur (1822–1896) used the microscope for his work on yeast and bacteria while, in 1876, Robert Koch discovered the microbes *Bacillus anthracis* – responsible for anthrax. For this Koch was awarded the 1905 Nobel Prize for Physiology and Medicine.

1 What is meant when something is studied with the 'naked eye'?

2 Draw up a time line showing important dates in the development of the microscope and discoveries made using microscopes.

3 Briefly describe the difference between a simple and a compound microscope.

4 What disease is caused by *Bacillus anthracis*?

5 What is a typical size for a bacterium cell?

Research

6 Walther Flemming (1843–1905) used microscopes in much of his work. Outline one of his discoveries in biology.

Match each branch of biology listed in the box to the correct definition below.

virology	*anatomy*	*pathology*	*mycology*	*ornithology*	*phycology*
cytology	*histology*	*embryology*			

7 Study of structure of plants and animals. _____

8 Study of fungi. _____

9 Study of plant and animal diseases. _____

10 Study of tissues (plant and animal). _____

11 Study of viruses. _____

12 Study of algae. _____

13 Study of early development of an organism. _____

14 Study of birds. _____

15 Study of structure and function of cells. _____

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Parts of the microscope

16 The compound light microscope used in school and university laboratories has changed little over the last 60 or so years. The boxes below show parts of a typical microscope and the functions of these parts. Cut out the boxes and match each part with its function. Stick them in your book under the headings 'Part' and 'Function'.

Ocular lens (eyepiece)	Lowest part, support for microscope
Used to get specimen approximately in focus	Has lenses of different magnification
Arm	Condenser lens
Stage	Pick up microscope by this part
Objective lens	Fine focus knob
Lens closest to specimen	Brings specimen into sharp focus
Revolving nosepiece	Microscope slide is placed on this
Mirror	Base
Reflects light onto specimen	Focuses light from mirror onto specimen
Place eye to look through this	Coarse focus knob

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